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Chi Yuan, Zhengbin Li, Yongqi He, Anshi Xu, "A novel node architecture for all-optical switching networks," Proc. SPIE 6784, Network Architectures, Management, and Applications V, 678430 (19 November 2007); doi: 10.1117/12.743623

SPIE.

Event: Asia-Pacific Optical Communications, 2007, Wuhan, China

A Novel Node Architecture for All-optical Switching Networks

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Abstract:

The explosion of Internet traffic has brought about an acute need for high-performance networks. The bandwidth available on a single fiber has increased dramatically by the wavelength division multiplexing (WDM) technologies. How to use the huge bandwidth flexibly and efficiently is one of the hot topics of optical networks. It is high time that the high-performance switches/routers should be designed. All-optical switching network is a networking platform with effective and agile utilization of the available optical bandwidth. Some promising optical switching networks, such as optical packet switched, optical burst switched, and optical label switched networks are proposed. The benefit of these networks compared to the existing optical WDM networks rises from the higher network utilization at sub-wavelength granularity and from the supporting of more various services. However, there are still a lot of challenges. One challenge of them is contention. Due to the fact that the viable optical random access memory is not available in the optical domain so far, there exists a real possibility that packets may contend with one another at a switching node. In this paper, a novel node architecture called packet calking switch is proposed for all-optical switching networks. The proposed architecture is accompanied by a signaling protocol for packet-based traffic handling. This proposed node architecture is introduced into a simple network and a mesh network through simulation. The simulation results show that it is suitable for prioritized packet transmission. And it is more cost-effective than the existing node architectures because it requires much smaller optical switches and wavelength conversion to achieve nearly the same packet drop performance as the node configured partial wavelength conversion.

Key words: All-optical Switching networks, Packet-calking Switch, Contention Resolution (CR), Wavelength Conversion (WC), Quality of Service (QoS)

I. Introduction

The explosion of Internet traffic, such as the continuous increasing peer-to-peer (P2P) traffic [1], has brought about an acute need for high-performance networks, especially the demand for high-performance switching networks. At the same time, the bandwidth available on a single fiber has increased dramatically by the wavelength division multiplexing (WDM) technologies [2]. How to use the huge bandwidth flexibly and efficiently is one of the hot topics of optical WDM networks. All-optical switching network is a networking platform with effective and agile utilization of the available optical bandwidth, which supports voice, data, and multi-media and new emerging applications. All-optical switching network is anticipated to address switching of asynchronously arriving variable-length switching entities (e.g. packets, bursts, and circuits) quickly in the optical domain. Some promising optical switching networks, such as optical burst switched (OBS) [3], optical packet switched (OPS) [4], and optical label switched (OLS) [5] networks are proposed to meet this requirement. These networks utilize optical switches to transparently transmit traffic through a network. The benefit of these networks compared to the existing optical WDM networks rises from the higher network utilization at sub-wavelength granularity and from the supporting of more various services. They can efficiently, flexibly and transparently transport bursty IP traffic over optical WDM networks.

However, for these all-optical switching networks aforementioned, if they want to be successful, there are still a lot of challenges. One of which is contention. Under the current and foreseeable limitations of optical technology, performance of an all-optical switching network is mainly hampered at the network nodes by resource contention. Due to the fact that the viable optical random access memory (RAM) is not available in the optical domain so far, there exists a real possibility that switching entities may contend with one another at a switching node. Contention will occur when two or more switching entities from different input ports wanting to

be forwarded on the same wavelength, on the same output port, at the same time. If not enough valid output ports for these switching entities, data dropping is inescapable in this instance. This feature is quite different from traditional packet-switched networks. Typically, contention in traditional electronic packet-switched networks is resolved through the store-and-forward technique, which requires the contending switching entity be stored in electronic RAM. Nevertheless, store-and-forward is not available in the optical domain at present for the absence of optical buffers.

Contention resolution in all-optical switching networks has attracted much attention recently. The most important design goal in these all-optical switching networks is to reduce data drop probability resulting from resource contention. To reduce contention probability, some methods have been proposed in literature. Existing work addresses this issue mostly focus on the wavelength domain (using wavelength conversion, WC [6]), the time domain (using fiber delay lines, FDLs [7]), and the space domains (using deflection routing [8]). These all-optical switching networks use wavelength converter and other methods rather than optical-electrical-optical conversion to achieve statistical wavelength multiplexing gains. Use of FDLs will increase data latency, and use of mass FDLs will increase implement complexity and is not commercially viable. Deflection routing will add data latency and traffic can loop in the network and waste resources in case of dropped. Although wavelength conversion is a good way jointed with other methods, but the technology is not mature at present. Therefore, it is high time that the high-performance switches/routers should be elaborated design. This study stems from practical observation that feasible limited WC is expensive and full WC is not viable at present.

In this paper, a novel node architecture called packet calking switch is proposed for all-optical switching networks. The proposed architecture is accompanied by a signaling protocol for packet-based traffic handling. Calking is the act or process of making seams tight. Packet calking mean filling gaps between packets by stuffing IP packets between adjacent nodes. With an orthogonal classification, all-optical switching networks (including OPS, OBS, and OLS, etc.) can be divided into two categories, namely, time slotted and unslotted networks. Both can have fixed-size or variable-size packets. Due to the fact that the viable optical RAM is not available so far, it is difficult to provide packet-level synchronization, which is required in time slotted networks. As for unslotted networks, it is suitable for variable-length packets without synchronization; it is also efficiently accommodate natural IP packets. Therefore, this paper primarily focuses on node architecture in unslotted OBS networks. This proposed node architecture is introduced into a simple network through simulation. The simulation results show that it is suitable for prioritized packet transmission. And it is more cost-effective than the existing node architectures because it requires much smaller optical switches and WC to achieve nearly the same packet drop performance as the node configured partial WC.

The rest of this paper is organized as follows. Section II introduces the contention in OBS networks. Section III introduces the Packet calking. Section IV presents the proposed node architecture. Section V evaluates the proposed node architecture through extensive simulations. Finally, we present our conclusions in Section VI.

II. Burst Contention and Resolutions in OBS Networks

In this section, we introduce the OBS network architecture in part A. Then we discuss burst contention and contention resolutions in OBS networks in part B.

A. Network Architecture

An OBS network is assumed that the network utilizes all-optical switching elements and ultra-long-reach all-optical transport. Basic components of an OBS network include some edge nodes, some core nodes, and some WDM links connecting these nodes together. Fig. 1 shows the OBS network architecture. The edge node is in charge of traffic aggregation, and the core node is in charge of burst switching and routing. The traffic from multiple access networks is accumulated at the ingress edge node and transmitted through high-capacity WDM links over the core node. The edge node provides legacy interfaces such as Gigabit Ethernet, ATM and IP interfaces, and are responsible for data burst assembly and disassembly. Data burst assembly refers to the process by which incoming packets are aggregated into data bursts according to some assembly algorithm. The reverse operation is referred to as the disassembly process. An ingress edge node is referred to the source edge node, and an egress edge node is referred to the destination edge node.

In an OBS network, incoming packets (such as IP packets) from the access networks are gathered and sorted at the ingress edge node according to their destination and quality-of-service (QoS). According to a preselect assembly algorithm, these packets are assembled into variable sized elementary switching and processing entities,

namely, burst data packet (BDP, throughout this paper, the terms BDP and burst are used interchangeably). Once a data burst has been created and ready to be transmitted by the edge node and based on the information of burst assembly and control information, a control packet (CP) will be generated synchronously by the edge node. To reserve network resource for its burst, a CP will be transmitted as soon as it is created. The burst is transmitted on dedicated data channels after an offset time. Commonly, the offset time includes basic offset time and QoS offset time. The former allows for the CP to be processed at the intermediate node in the electrical domain and the network resource reservation, and the latter allows for the burst be transmitted with differentiated QoS. All CPs are transmission on the control channel, which may be separated from the data channels. All CPs are processed by intermediate core nodes in the electrical domain. According to the processing information (such as burst length, offset-time and QoS, etc.), the intermediate core node reserves network resources orderly for this burst's transmission. The burst is switched all-optically without waiting for acknowledgment of light path setup [3, 9-10]. It follows its CP and cut-through the OBS network without optical-electrical-optical conversion.

B. Contention and Contention Resolution

OBS is designed to avoid the long end-to-end setup time of conventional virtual circuit configuration with no need for buffers at the core node. Owing to the fact that optical logic and RAM-like buffering is not available in the optical domain, transmission of bursts will fail when network resource reservation fails for two or more bursts competing for the same resource simultaneously, knowing as burst contention. That is, contention will occur when two or more bursts from different input ports wanting to be forwarded on the same wavelength, on the same output port, at the same time. Contention is inherent to OBS technology, which basically assumes that the network is buffer-less and connectionless. This feature is quite different from traditional packet switching networks. Typically, contention in traditional electronic packet switching networks is handled through buffering. However store-and-forward is not available in OBS networks. To avoid burst dropping, some contention resolution methods have been proposed in the literature.

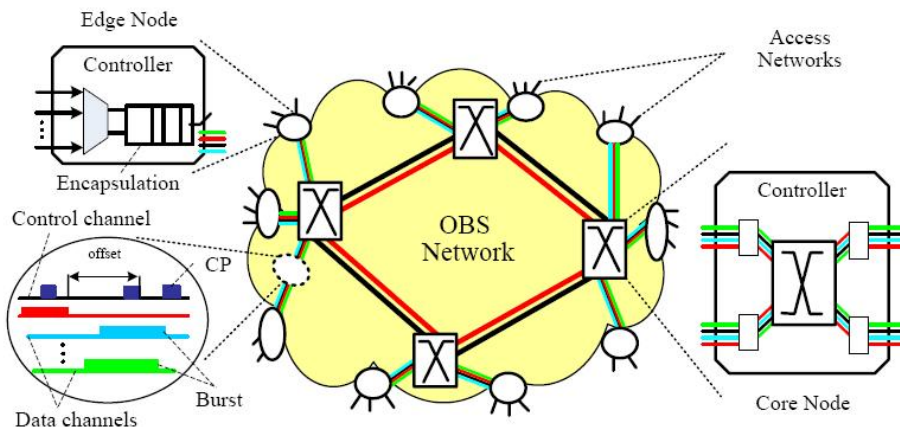


Fig. 1 Optical burst switched network architecture

Obviously, most contention resolution methods in the literature so far can be classified as reactive and proactive approaches. Reactive approaches are provoked after contention occurs. Examples of reactive contention resolution schemes are the space domain (as deflection routing [8]), the time domain (buffering by using FDLs [11]), the wavelength domain (as wavelength conversion [11]), the burst domain (as segmentation [12]), and the control domain (as look-ahead contention resolution [13] and shortest-drop policy [14]). Although wavelength conversion is a good way jointed with optical buffering, but the technology is not mature at present. Use of FDLs will increase data latency and implement complexity and is not commercially viable. Segmentation will increase complexity of control and implement action. Deflection routing will add burst latency and a burst can loop in the network and waste resources in case of dropped burst and invalidation.

In proactive contention resolution approaches, traffic management policies are invoked to prevent the network entering the congestion state. Such schemes can be classified as non feedback-based and feedback-based. In a non feedback-based scheme, the ingress edge nodes have no knowledge of the network state and they cannot respond to changes of the network load. This can be achieved through traffic load balancing or data burst assembly. In a feedback-based scheme, contention avoidance is achieved by dynamically varying the data burst

flows at the ingress edge node to match the latest status of the network and its available resources. One way to achieve this is to reroute some of the traffic from heavily loaded paths to under-utilized paths [15]. A similar approach has also been introduced by [16] where the authors consider balancing the data burst traffic between predefined alternative paths. In [17] a global load-balancing contention resolution scheme is proposed and its performance is examined for both dynamic and static traffic.

Another way to avoid contention is to implement a burst transfer control protocol (BTCP) with a congestion avoidance mechanism to regulate the burst transmission rate [18-20]. In this approach, the ingress edge node receives BTCP acknowledgment (ACK) packets from the egress edge node, calculate the most congested links, and reroute their traffic accordingly. In [21], burst copying is proposed as an effective way to reduce data burst loss. The basic idea in burst copying is to replicate a burst at an appropriate node and send duplicated copies of the burst through the network simultaneously. In [22], a feedback-based OBS network is proposed by using explicit feedback signaling to each source, the required data burst flow rate going to congested links is controlled. Clearly, a major concern with having feedback-based proactive contention resolution schemes is additional signaling overhead and control message processing. Hence, it is critical to design signaling protocols which are simple to implement and require minimum overhead.

Reference [11] shows an ideal network, where the core node uses a number of FDLs in the time domain and uses wavelength converter in the wavelength domain to reduce contention. But contention still occurs at a switching node when the load gets higher. It indicates that these methods may reduce the burst contention probability, but they are very vulnerable to network load and may suffer from severe data loss in case of heavy traffic load. Even at moderate traffic load, contention caused by using these methods lead to burst blocking and data drop probabilities have a long way to go to commercially viable. Undoubtedly, a combination of mentioned above methods can be very effective. Therefore, to improve the performance of OBS networks, we proposed a novel burst scheduling algorithm. This unique contention resolution approach can be used in conjunction with a differentiated Quality of Service (QoS) scheme to reduce the burst drop probability.

III. Packet Calking

Longer burst produces less total number of burst injected into the OBS network than what the smaller burst does. But in the case of contention, the lost packet per contention will increase. Whereas smaller burst increase the total number of burst more than that of longer burst, this will lead to higher contention probability. If OBS can transmit some packets on demand according to the wavelength engagement, moreover, if OBS can fulfill the absurdity that reducing total number of the burst and not increasing the mean length of them, the network would get less contention. Therefore, we propose a mechanism that the single-hop traffic which is not assembling into burst. The ingress node monitors and transmits these single-hop IP packets at an appropriate time. This mechanism is illustrated in a four-node network as Fig. 2. Traffic between nodes 0 and 1, nodes 1 and 2, nodes 2 and 3 are single-hop traffics. These traffics are not assembling into burst. They are stored at electronic buffer of the ingress node and are transmitted directly between adjacent nodes. This mechanism has two advantages to the OBS network due to the single-hop traffic not assembling into burst. One advantage is that there is no assembly delay for single-hop traffic. The other advantage is that the burst number could be reduced largely.

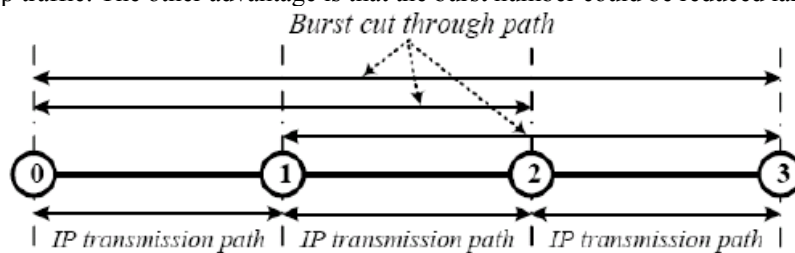


Fig. 2 The transmission mechanism of traffic

Packet calking based on the two advantages mentioned above. Calking is the act or process of making seams tight [23]. Packet calking mean filling gaps between packets by stuffing IP packets between adjacent nodes. The basic function of a packet calking switch is described as follows: (a) the node categorizes the adding network traffic into two types--single-hop and multihop traffic, based on the fact that it uses one or more physical links; (b) the node supports prioritized packet transmission, and the pass-by multihop traffic is always forwarded first; (c) the new adding traffic can be aggregated if needed; and (d) single-hop traffic be calked into gaps of multihop

traffic packets to transmit.

In an OBS network with packet calking, the node transmits both the multihop burst and the single-hop IP packets. From the perspective of a node, the single-hop traffic is stored at electronic RAMs of the node. If it detects that a wavelength is idle in a specifically period, then some single-hop IP packets are aligned as a calking entity which has a proper duration to fit the period. The node calks and transmits the entity to the neighboring node. In Fig. 2, traffic between nodes 0 and 2, nodes 0 and 3, nodes 1 and 3 are multihop traffic, which is assembled into bursts according to a preselected assembly algorithm. The node stores the burst and sends a CP to book network resource. This burst is transmitted after an offset time, and it cuts-through the path from the ingress node to the egress node.

IV. Node Architecture

In the architecture of aforementioned optical switching networks, IP traffic entering the network is processed by an edge node at the edge/core interface, where it is aggregated into different switching entity with various granularities (e.g. flow, burst and packet) that are delivered to the core optical layer. To get high switch efficiency with an affordable delay, OPS need traffic aggregating. The all-optical sub-wavelength switching technologies with statistical multiplexing poses problems that are more similar to those faced in legacy packet-switched networks, such as resource allocation and congestion resolution, but again the problem has some peculiar characteristics due to the optical technology. So the node architectures require large optical switches and WC.

This proposed node is assumed to include an all-optical switch fabric to add, drop and/or switch the traffic at various granularities, as well as to provide optical bypass for traffic that does not need to drop at the node. The proposed node architecture for OBS networks is shown in Fig. 3. It is buildup of a traffic sorter, a burst assemble/disassemble module, an assemble control module, an IP packet transmit /receive module, some electronic random access memory (RAM), a transmission scheduler, a transmitter (T_x) and a receiver (R_x) module, an optical switch fabric, and a wavelength division multiplexing (WDM) multiplexer marked 1 and a demultiplexer marked 2.

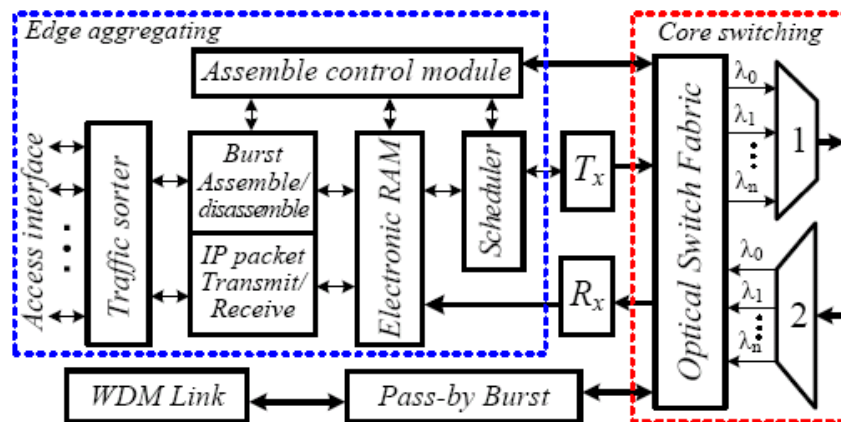


Fig. 3 Function modules of an OBS node

The node has two important function parts: edge aggregating and core switching. The former is in charge of traffic aggregation. The latter is in charge of traffic switching. IP packets from the access interface are sorted by the traffic sorter according to their destinations. If it is the adjacent node, then the packet is queued in the single-hop queue, else in the multihop queue. IP packet of the multihop queue is assembled into burst according to a preselected burst assembly algorithm. Both the burst and the single-hop IP packet are stored in the electronic RAM. If the node detects that a wavelength is idle in a specifically period, then some single-hop IP packets are aligned as a calking entity which has a proper duration to fit the period.

Before the node transmits a calking entity which destined for an adjacent node, the node sends a calking control packet which is shown in Fig. 4 to inform the adjacent node. There is a signaling protocol for the adjacent node pair to transmit and receive a calking entity. The rule of the packet transmission is that the by-pass burst should be cut-through first, and the adding burst and calking entity should allocate bandwidth second. Both entity and burst are switched all-optically.

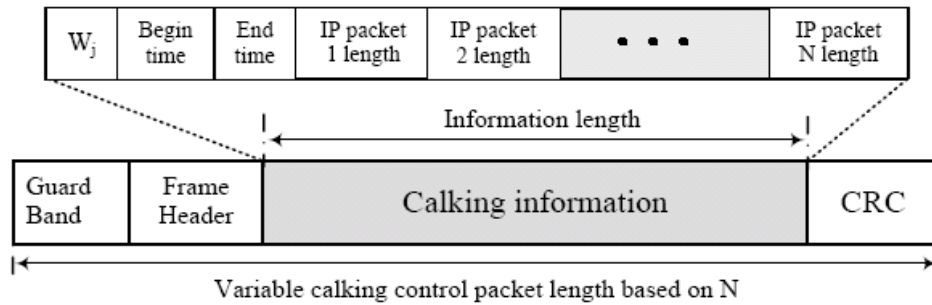


Fig. 4 Calking control packet format

The optical switch fabric is shown in Fig. 5. There are N input and N output fibers, and each fiber has $m+1$ wavelengths. It is assumed that λ_0 is dedicated to control channel, $\lambda_1, \lambda_2, \dots, \lambda_m$ are used as data channels. The optical switch fabric is control by a switch control unit. This figure traces out the switch process of four pass-by bursts which marked 1 to 4 and two calking entities which marked C_1 and C_2 respectively. When the node find a gap between burst 1 and burst 2 which destine for the λ_m port of output fiber N , it calks the calking entity C_1 into this gap.

V. Performance Evaluation

To evaluate the proposed node architecture, five scenarios, namely packet calking, no calking, FDLs, partial WC (PWC), and packet calking combined PWC, are developed in a simple network as shown in Fig. 1, and both packet calking and no calking scenarios are developed in the 14-node 21-link National Science Foundation Network (NSFnet) in OPNET simulator. There is a bottleneck link between the middle nodes in the simple network. The link has 4 data channels and each at 10 Gbps. In FDLs scenario, the node configured two FDLs to resolve contention. In PWC scenario, the node has configured PWC. That is λ_1, λ_2 could be converted each other and λ_3, λ_4 as well. But λ_1 and λ_2 couldn't convert into λ_3 or λ_4 , and vice versa. An assembly algorithm as literature [24] is selected and the following assumptions are made for the simulations:

- IP packets arrival is a Poisson process
- IP Packet's length is exponential distribution
- The average IP packet length is 1250 Bytes
- The Offset time is set 0.5 milliseconds
- All nodes employ first-in first-out scheduling policy
- Switching & processing time is 5 microseconds
- Just enough time (JET) [3] OBS based signaling

Fig. 6 shows the packet drop probability versus offered load. It can be easily proved that the proposed node architecture requires much smaller optical switches and WC than the PWC node architecture. But it gives the same drop performance as PWC when offered load less than 0.3. When load excess 0.3, the packet calking scenario gets better drop performance. Packet calking combined PWC scenario gets the best drop performance among the five scenarios. It gets about 25~30% packet drop probability of no calking scenario. Fig. 7 demonstrates that the bottleneck and ordinary link utilization versus offered load. The curve indicates that packet calking achieves a little higher utilization than that of no calking when offered load is less than 0.3. When offered load is between 0.3 and 0.5, link utilization of calking is 10% higher than that of no calking. When load excess 0.6, link utilization of calking is 25% higher than that of no calking.

Fig. 8 plots the burst drop probability of packet calking and no calking scenarios in the NSFnet respectively. The 14-node NSFnet is an irregular topology network. From Fig. 8, we can see the IP calking outperforms than no calking scenario throughout the load range considered in this figure. These curves indicate that the packet calking switch achieves a significant lower burst drop probability. The burst drop probability of packet calking is about 30% lower than that of no calking. Fig. 9 shows the end to end (ETE) delay of burst and IP packet in 14-node NSFnet. In this figure, the ETE delay of IP is not obviously increased with increasing load. The ETE delay of burst is decreased with the increasing load. The reason is that the data-length time-lag product assembly algorithm, which using product of aggregated data length and assembly time-lag as a threshold. It is considered the aggregated data length and the assembly time-lag simultaneously. The assembly time-lag is decreased with increasing load at a given threshold. Therefore, the ETE delay of bursts decreases with increasing load.

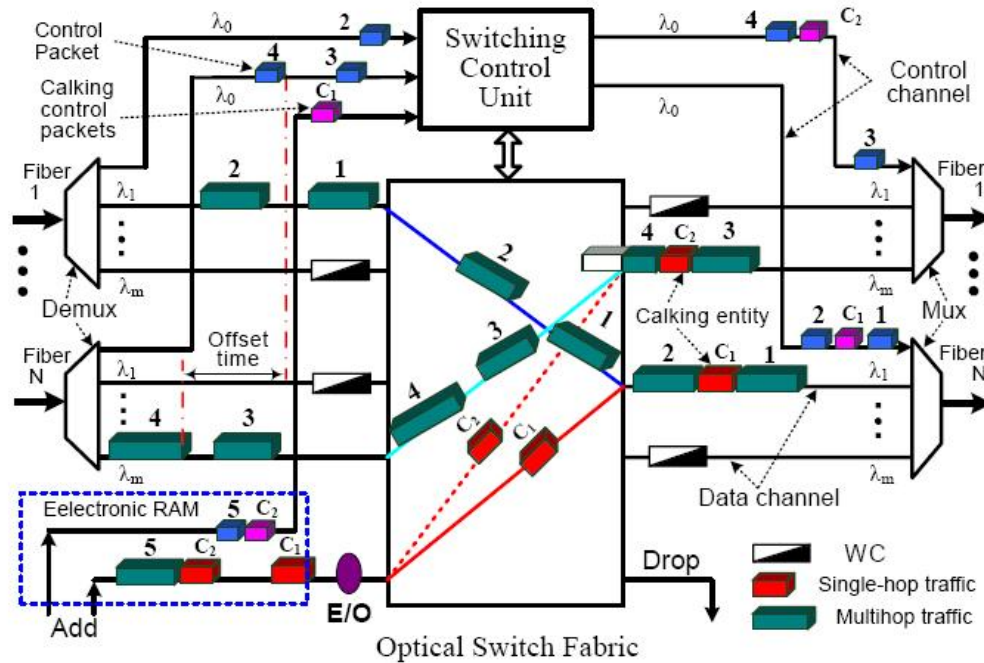


Fig. 5 Architecture of an optical switch fabric

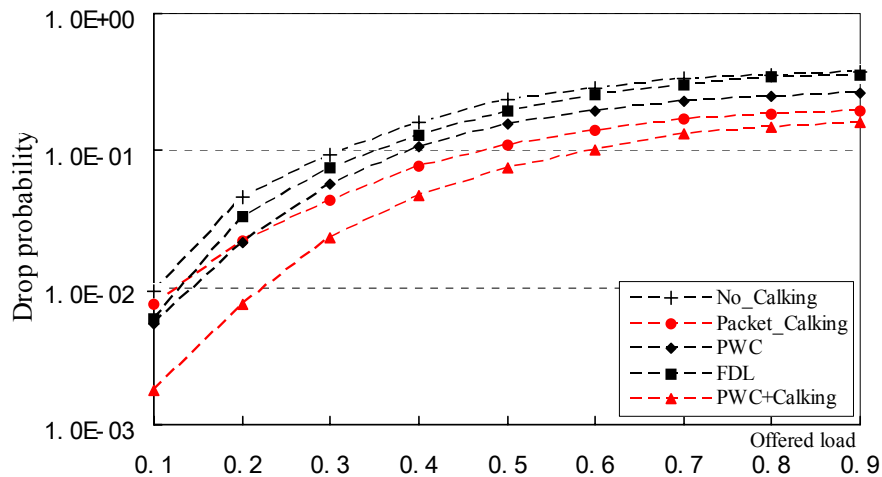


Fig. 6 Numerical drop probabilities versus offered load

VI. Conclusion

In this paper, a new node architecture called packet calking switch is proposed for all-optical switching networks. The proposed node architecture is suitable for prioritized packet transmission. We take the OBS network as an example of all-optical switching network. The proposed node architecture is introduced in a 4-node line network and the 14-node 21-link NSFnet. The burst drop probability, ETE delay, as well as link utilization were attained. The simulation results show that the proposed node architecture outperforms than the traditional node architecture, without reference to regular line network or irregular mesh network. It is more cost-effective than the existing node architectures because it requires much smaller optical switches and WCs to achieve nearly the same packet drop performance as the node configured PWCs.

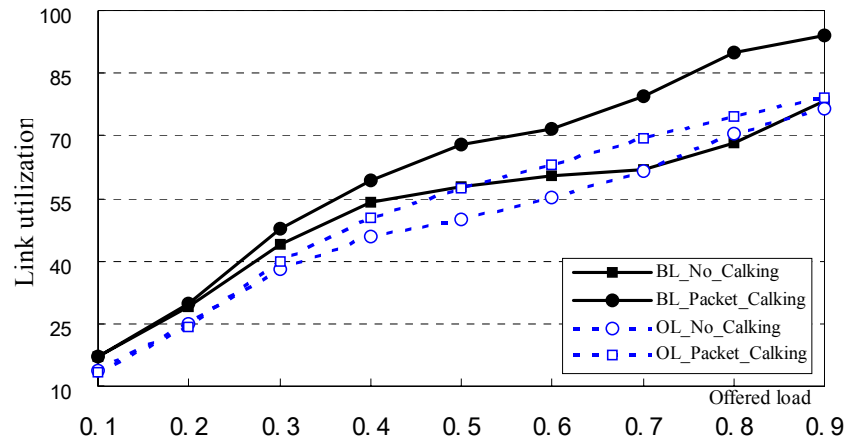


Fig. 7 Link utilizations versus offered load

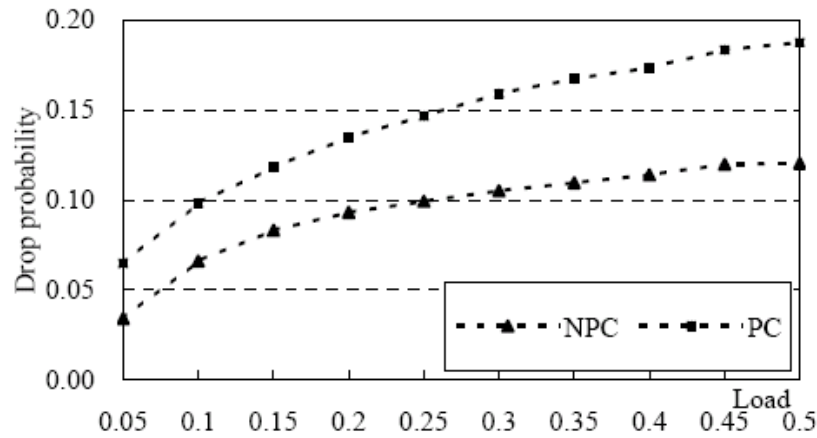


Fig. 8 Burst drop probability varying with load

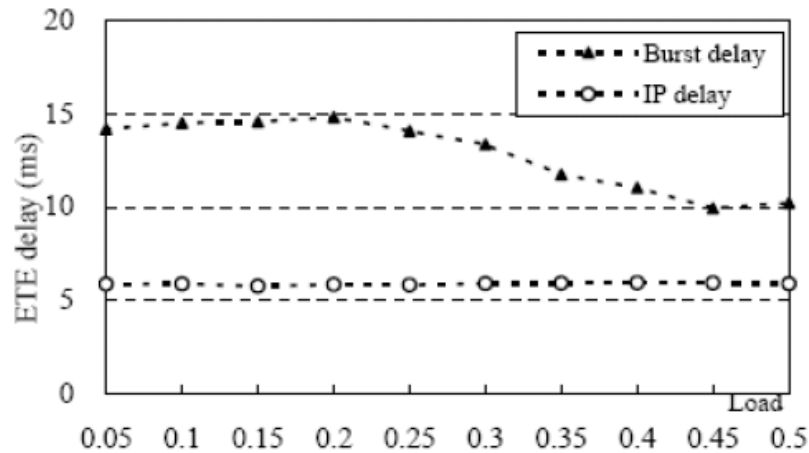


Fig. 9 ETE delay of Burst and IP packet in 14-node NSFnet

Acknowledgment

This work was supported by National Hi-tech Research and Development Program of China (863 Program) (No. 2006AA01z249), Project of Ministry of Education of China (No. 105002), and National Natural Scientific Foundation of China No. 60372025.

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